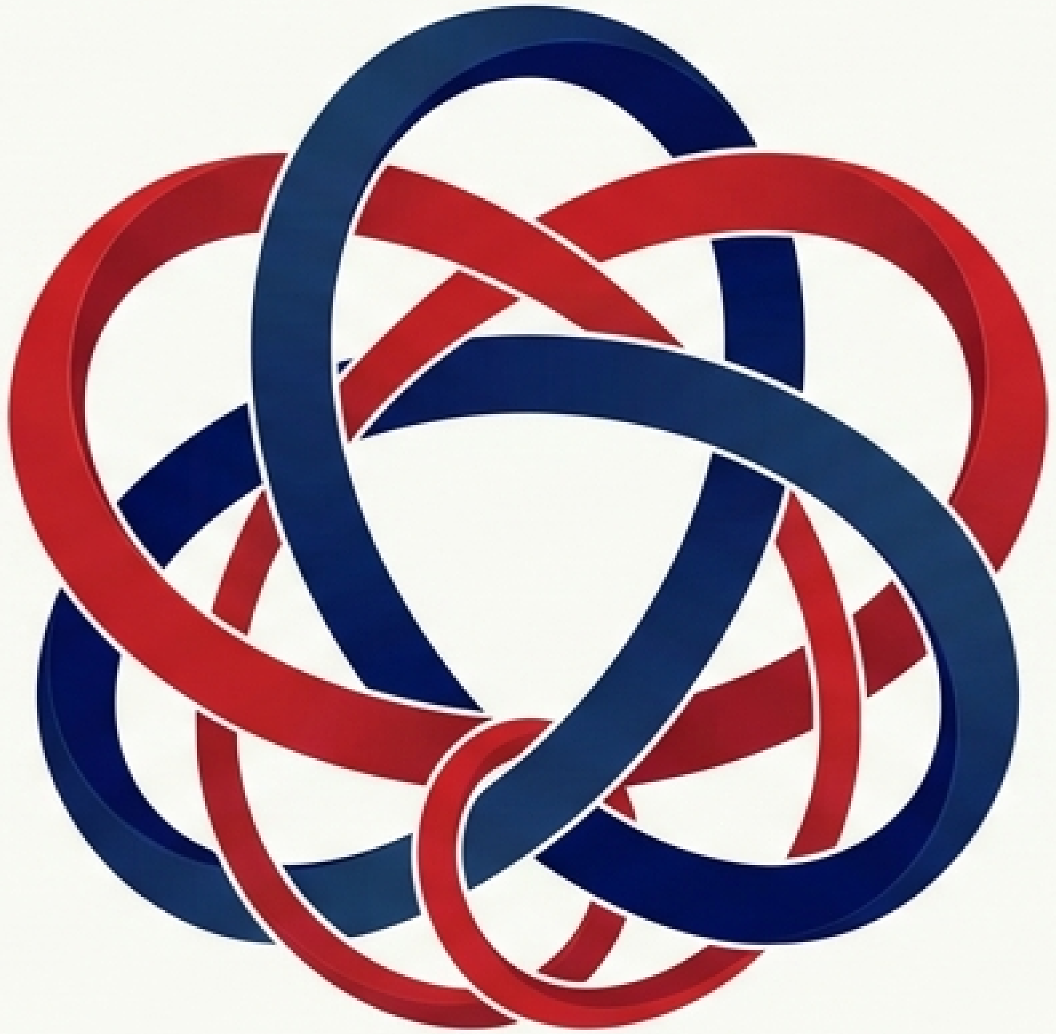
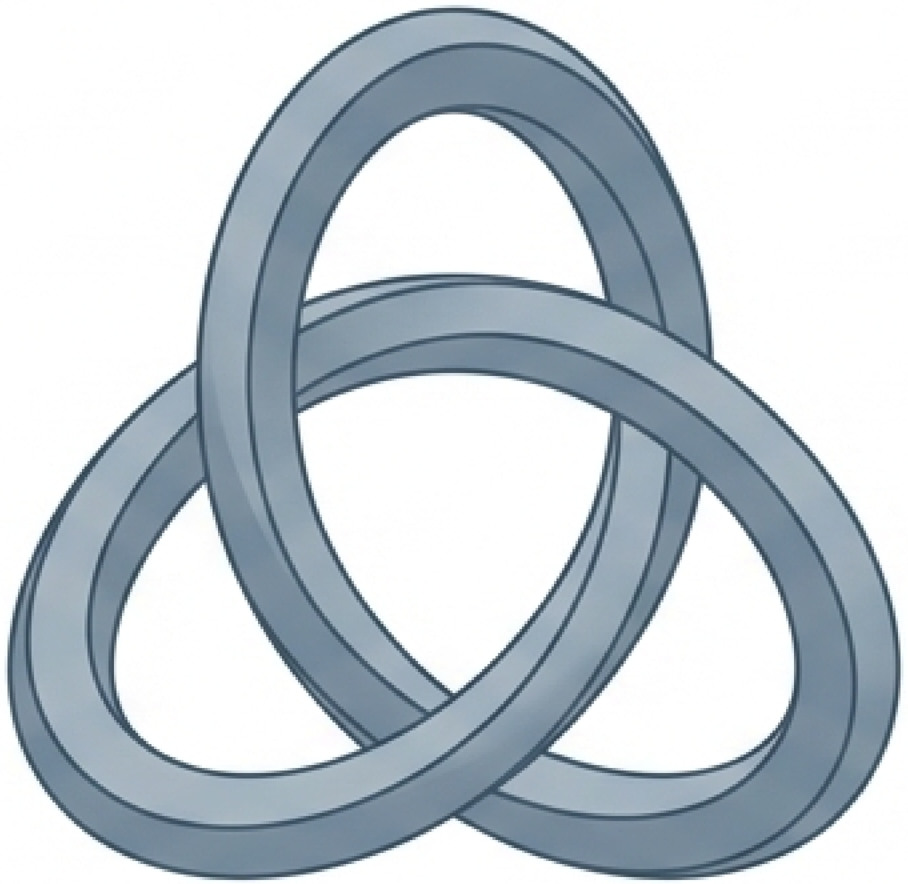
Atomic Masses from Topological Invariants



A Classical, Geometry-Driven Framework for Elementary & Nuclear Mass

BASED ON THE RESEARCH 'SST-59' BY OMAR ISKANDARANI (JANUARY 2026)



CORE THESIS:

Mass is not a quantum interaction parameter.

Mass is a geometric invariant of knotted fields.

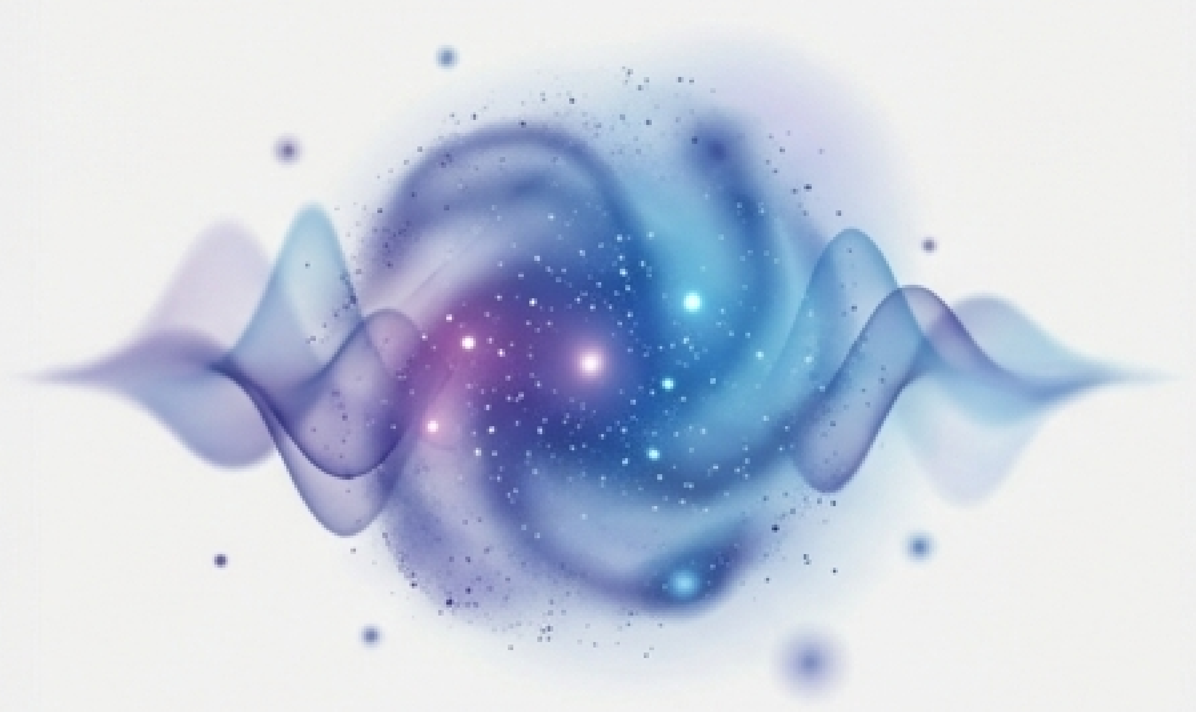
PREDICTIVE ACCURACY

Particle	Configuration	Relative Error
Electron	Trefoil	0.000%
Proton	Link (n = 3)	0.000%
Neutron	Link (n = 3)	0.000%
Uranium	Complex	$< 0.7\%$

Derivation independent of Higgs couplings or quantum postulates.

A Shift from Quantum Interaction to Geometric Structure

STANDARD APPROACH



PROBABILISTIC DYNAMICS

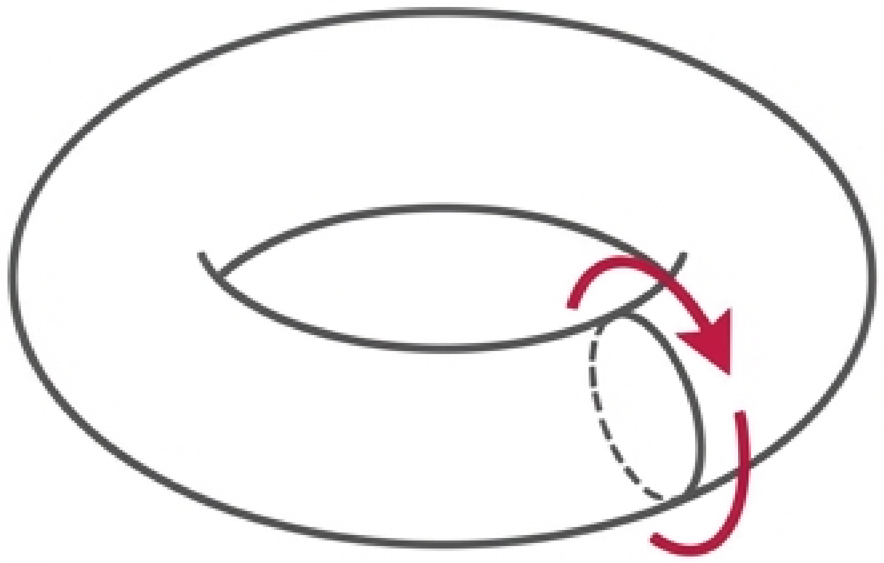
- Relies on Quantum Electrodynamics (QED)
- Mass via Higgs Coupling
- Interaction-dependent parameters

TOPOLOGICAL APPROACH

DETERMINISTIC GEOMETRY

- Relies on Classical Field Theory
- Mass via Geometric Invariants
- Structural properties only:
  - k (Kernel Suppression)
  - g (Genus)
  - n (Components)
  - L_{tot} (Ropelength)

Foundation I: Canonical Energy Density



$$u = \frac{1}{2}\rho v_{\circlearrowright}^2$$

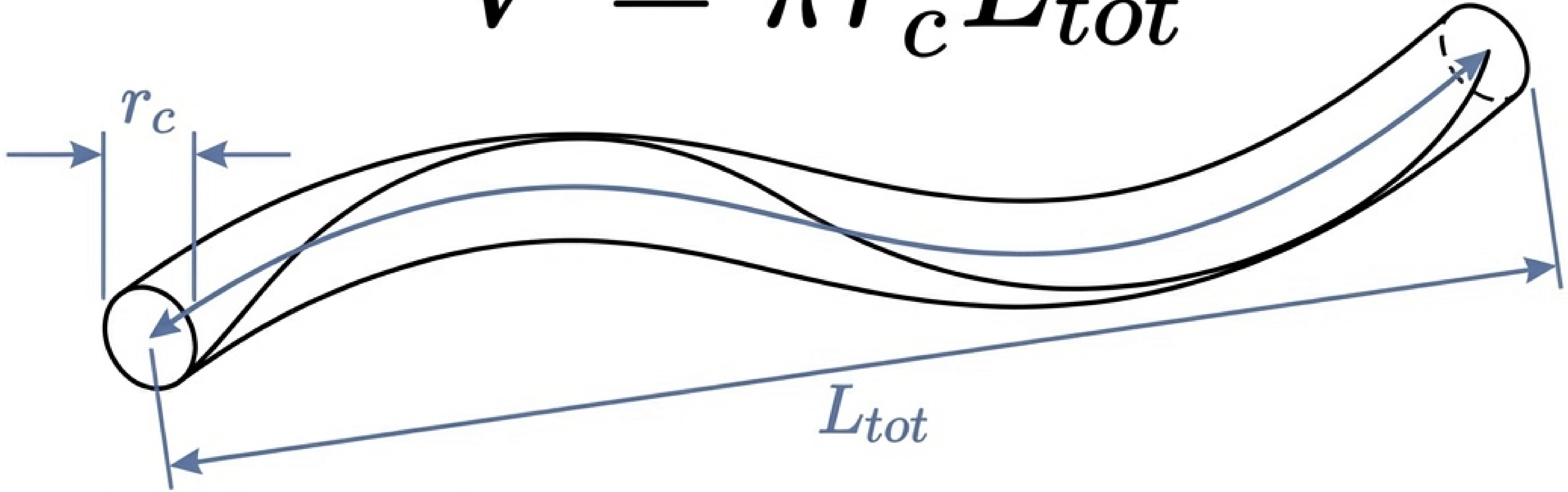
The ‘Substance’ of Mass:

- u: Classical Kinetic Energy Density (J/m^3)
- ρ: Effective core density (constant)
- $v_{\circlearrowright}$: Characteristic tangential circulation velocity

This sets the universal energy scale. No quantum fluctuations.

Foundation II: The Geometric Volume

$$V = \pi r_c^3 L_{tot}$$



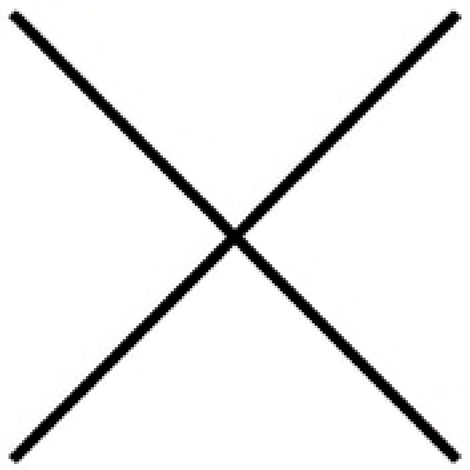
The 'Shape' of Mass:

- r_c: Core Radius. Defines the transverse cutoff / coherence length.
- L_{tot}: Dimensionless Ropelength. Pure geometry.

All dimensional scaling is locked into the radius r_c.

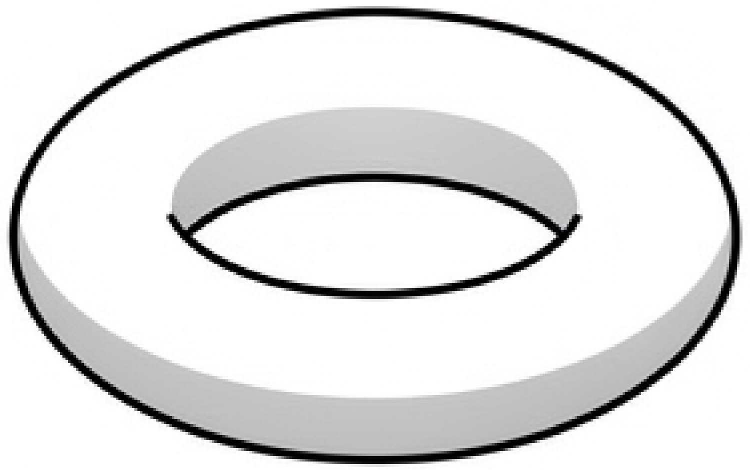
The Core Logic: Topological Suppression

Increasing geometric complexity suppresses energy localization efficiency.



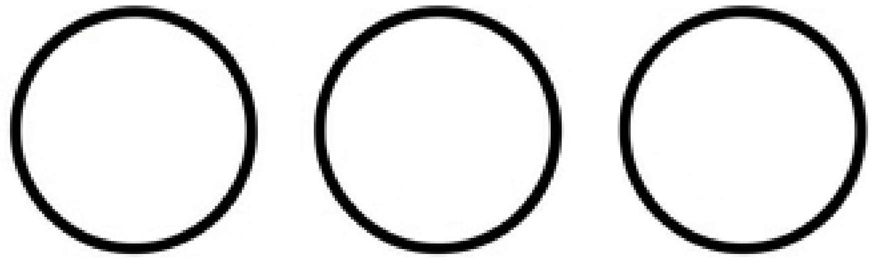
Kernel Suppression (k)

$$S_k = k^{-3/2}$$



Seifert Genus (g)

$$S_g = \exp(-g \cdot \operatorname{asinh}(1/2))$$



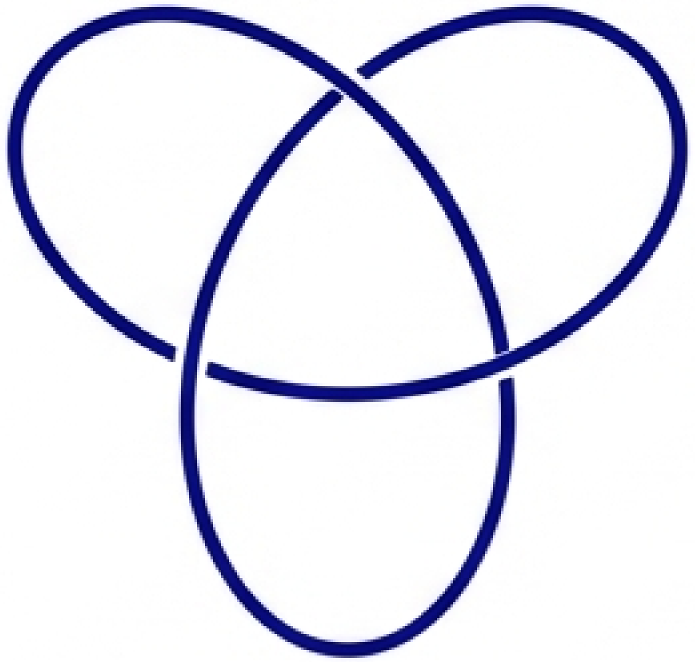
Component Number (n)

$$S_n = n^{-\exp(-\operatorname{asinh}(1/2))}$$

Note: The constant $\operatorname{asinh}(1/2)$ is a fixed geometric value arising from logarithmic scaling, not an arbitrary tuning parameter.

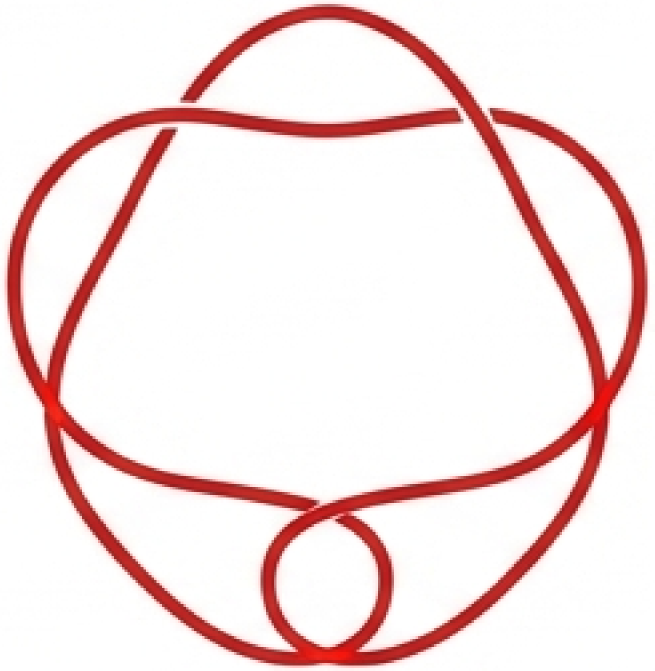
Visualizing the Topology

Elementary particles correspond to specific knot classes.



Electron (e^-)

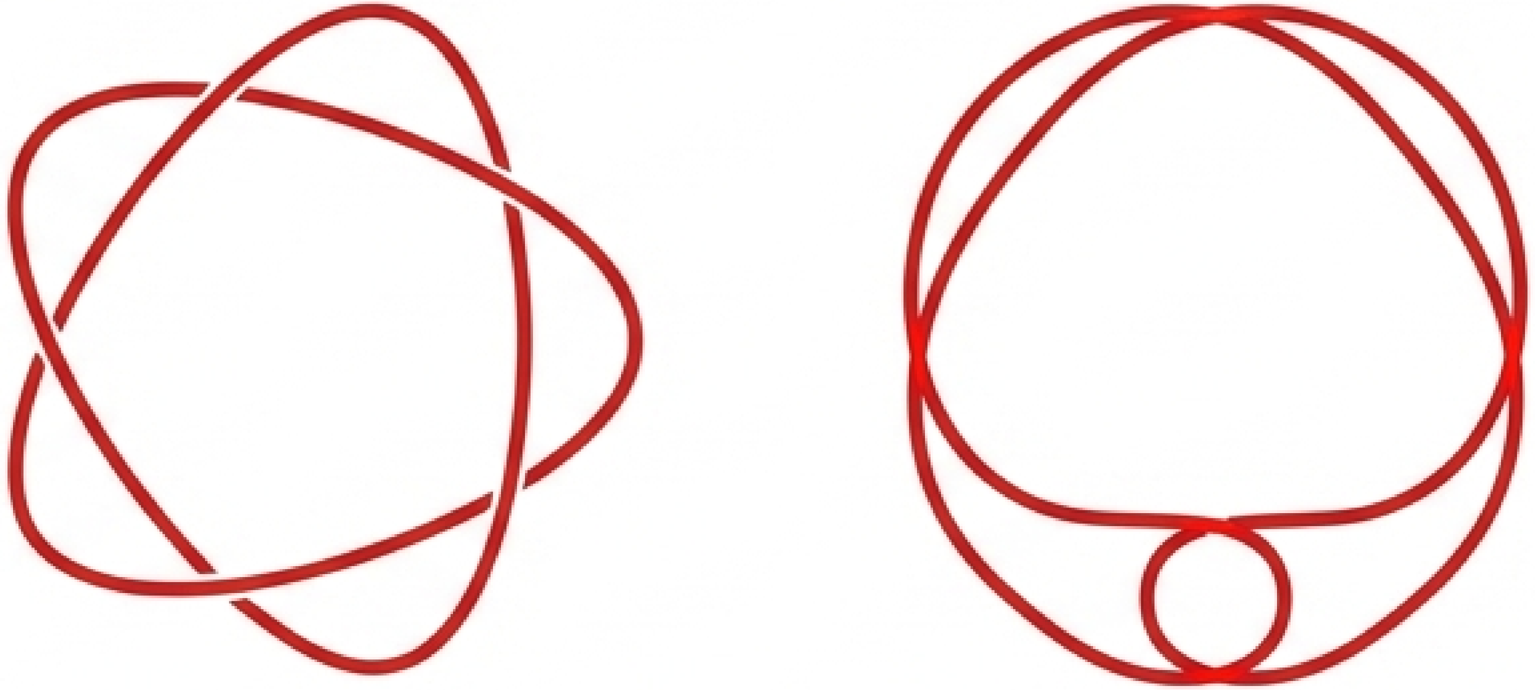
Trefoil Knot 3_1 (Chiral)



Muon (μ)

Cinquefoil Knot 5_1

Quark Representatives



Quark Representatives

Twist Knots 5_2 and 6_1

Achiral knots are excluded as inert, non-localizable energy modes.

The Invariant Master Equation

$$M(T) = \left(\frac{4}{\alpha} K(T)\right) \cdot \left(\frac{uV}{c^2}\right)$$

The Fine Structure Constant appears as a universal amplification factor.

Expanded Form

$$M = \underbrace{\frac{4}{\alpha}}_{\text{Amplification}} \cdot \underbrace{\left[k^{-3/2} e^{-g\cdots} n^{-\cdots}\right]}_{\text{Topology (K)}} \cdot \underbrace{\left[\frac{1}{2}\rho v^2 \cdot \pi r_c^3 L_{tot}\right] / c^2}_{\text{Energy \& Volume}}$$

Code Isomorphism: From Math to Python

$$M(T) = \left(\frac{4}{\alpha}K(T)\right) \cdot \left(\frac{uV}{c^2}\right) \qquad (10)$$

```python
# Exact implementation of Equation (10)
total_mass = (
    amplification * # 4 / alpha
    kernel_suppression * # k term
    genus_suppression * # g term
    component_suppression * # n term
    (u * volume) / (c * c)  # Energy/Volume term
)
```

No hidden physics. The theory is implemented verbatim in the reference code.

Calibration Strategy: The Electron Standard

We do not fit the curve; we fix the scale.

Input
Measured Electron Mass (M_e)

Invert Eq. (10)

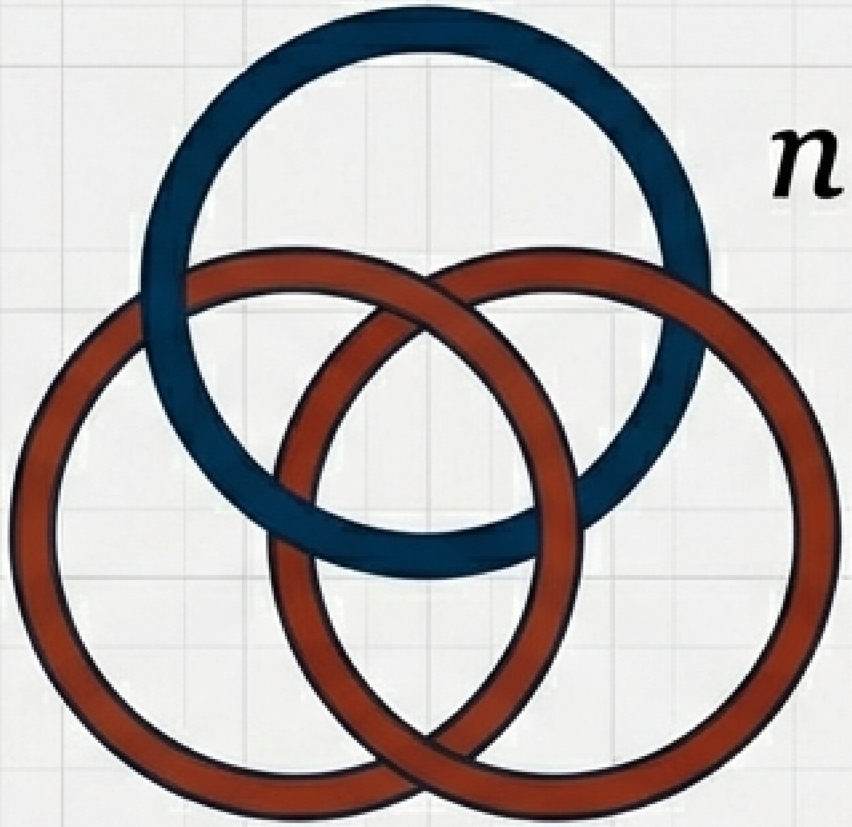
Calibration
Solve for Electron Ropelength (L_{tot})

Fixes Scale for Universe

Output
Predict All Other Particle & Atomic Masses

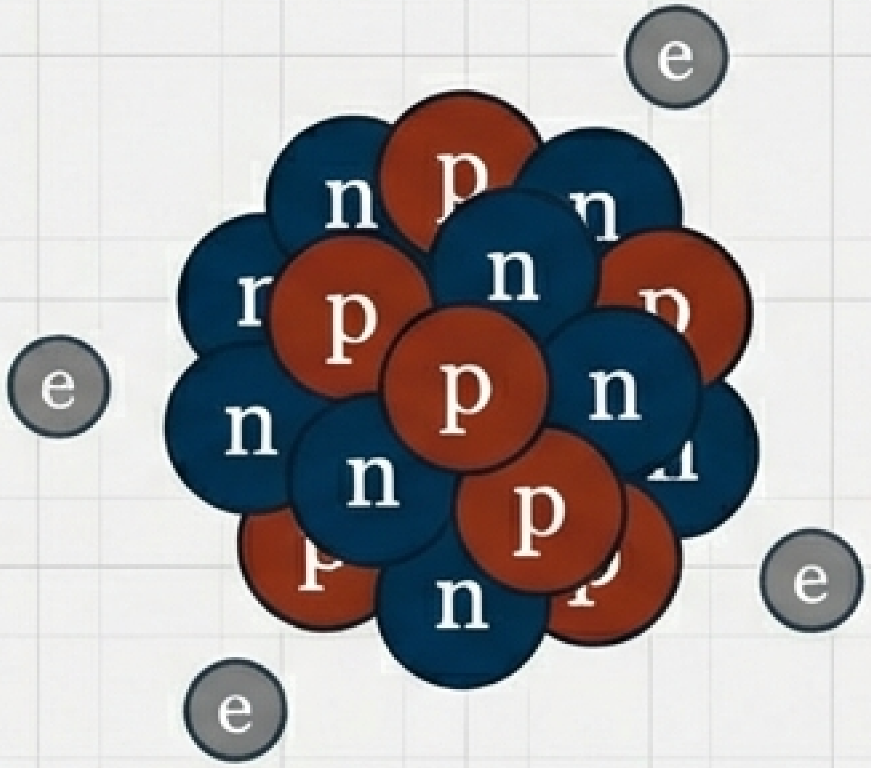
Once the electron ropelength is fixed, the logic runs autonomously. No per-element fitting.

Scaling Up: Baryons and Nuclei



modeled as 3-component link configurations.

$$L_{tot} = \sum L_{parts}$$



Hybrid Approximation for Atoms:

$$M_{atom} = ZM_p + NM_n + ZM_e - (E_{bind}/c^2)$$

Constituent masses derived from Topology. Binding energy (E_{bind}) from Standard Nuclear Model.

Evidence I: Elementary Precision

Particle	Actual Mass (kg)	Predicted Mass (kg)	Error %
Electron (e^-)	9.109×10^{-31}	9.109×10^{-31}	0.000%
Proton (p^+)	1.672×10^{-27}	1.672×10^{-27}	0.000%
Neutron (n^0)	1.674×10^{-27}	1.674×10^{-27}	0.000%

"Precision is not due to overfitting, but a stable invariant geometric structure."

Evidence II: Global Consistency (Atoms)

Residuals track known secondary nuclear structure effects (shells, deformation) outside the topological approximation.

Dimensional Consistency

$$\underbrace{[J \cdot \cancel{m}^{-3}]}_{\text{Energy Density}} \times \underbrace{[\cancel{m}^3]}_{\text{Volume}} = [J]$$

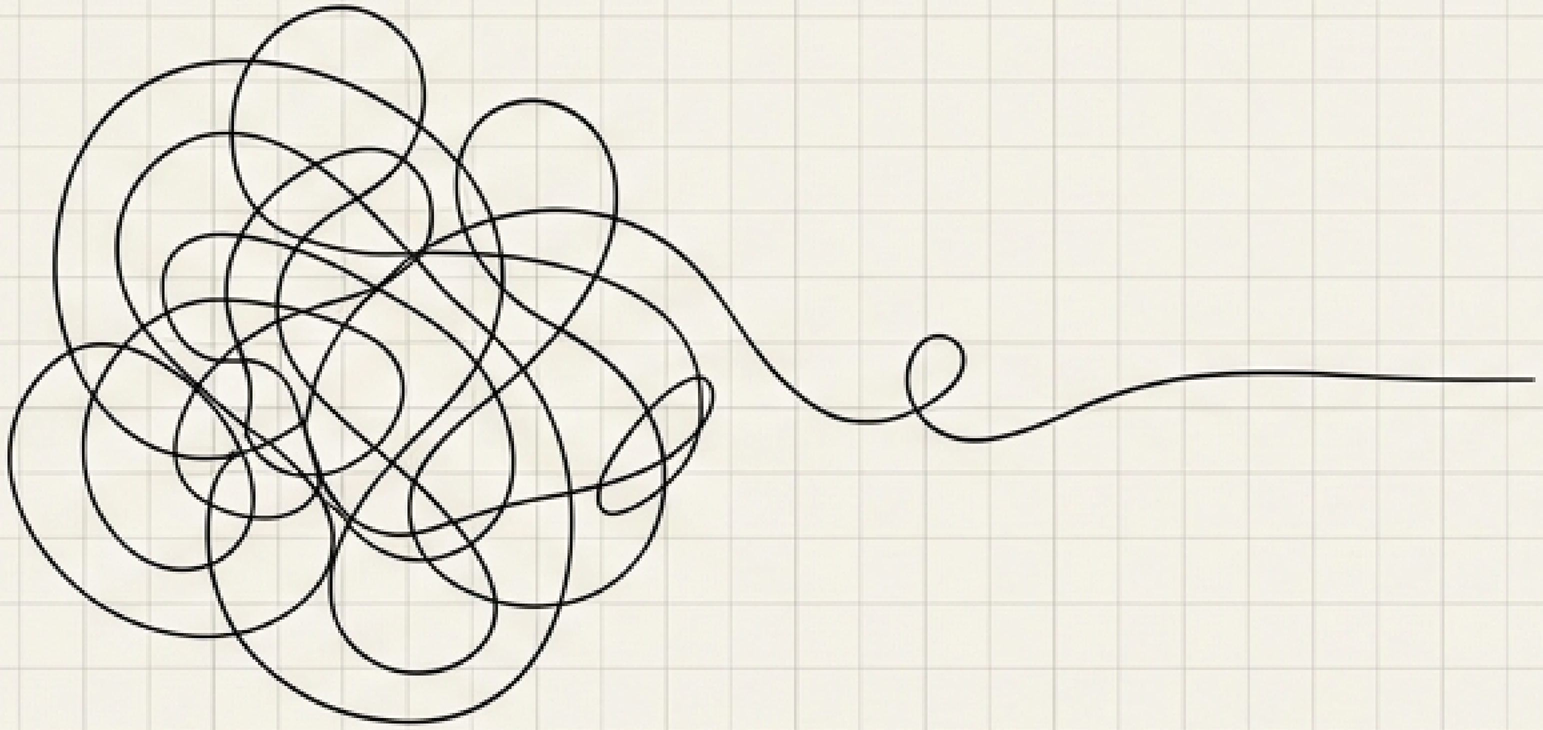
Dimensionless Terms:

- α (Fine Structure)
- k, g, n (Topology)
- L_{tot} (Geometry)

$$\frac{[J]}{[(\mathrm{m/s})^2]} = \frac{[\mathrm{kg} \cdot \cancel{\mathrm{m}^2}/\cancel{s^2}]}{[\cancel{\mathrm{m}^2}/\cancel{s^2}]} = \mathbf{[kg]}$$

Result: The formula is dimensionally sound and outputs Kilograms.

Conclusion: The Geometry of Mass



- Exact Analytical Counterpart: Code and Math are isomorphic.
- No Speculation: No hidden parameters or auxiliary dynamics.
- The Discovery: Mass is encoded in a geometric-topological invariant kernel.

The *mass* of the universe may be less about the interaction of particles, and more about the shape of the field itself.